Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/09/2005
PI Name:	Cardoso Landa, Luis Ph.D.		
Project Title:	Ultrasonic Assessment of Anisotropic Mechanical Properties of Cancellous Bone after Disuse with and without Anti-Resorptive Therapy		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10029-6504	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSBRI-RFP-04-01 Postdoctoral Fellowships
Start Date:	11/01/2004	**End Date:**	10/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: project ended after one year, per K. Major (3/2007)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schaffler, Mitchell (MENTOR: Mount Sinai School of Medicine)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Currently, the approach most widely used to examine bone loss is bone densitometry, which cannot show the microarchitectural aspects of cancellous bone that are key to estimate bone's mechanical integrity. Recently, a novel architecture-density based approach for acoustic wave propagation in a poroelastic media was developed. This approach provides the potential to use ultrasound measurements in bone to examine tissue architecture in addition to bone mass. Initial studies show that this approach significantly improves prediction of mechanical properties of bovine and human cancellous bone when compared to density-based approaches. Better estimation of bone mechanical properties is expected to result in enhanced discrimination of osteoporotic and non-osteoporotic bone. The Aim of the present study is to investigate the ability of this microstructural-based approach to discriminate quantitatively the changes on density, microstructure and mechanical properties of bone in the real-world situations of a rapidly evolving osteoporosis (disuse), both with and without antiresorptive treatment, and after recovery. The present research builds on the ongoing NSBRI studies (NSBRI BL00203 (Resorption Suppression and Bone Health in Disuse)) conducted by my mentor Dr. Mitchell Schaffler at the Orthopaedics department in Mount Sinai School of medicine. From these studies, canine cancellous bone samples were obtained and split into four groups: Immobilized (Im), Immobilized and treated with antiresorptive therapy (ImTx), Non-immobilized control bones (Cn), and non-immobilized control bones treated with bisphononate (CnTx). Ultrasound wave velocities were measured in three anatomical directions (SI, ML and AP). The values of ultrasound velocities and the bone density were used to directly compute the directional anisotropic mechanical properties of cancellous bones in the four analyzed groups. In particular, from these measurements it was found a noticeably change in mechanical properties when comparing control and immobilized cancellous bone samples. The group of bones immobilized and treated with antiresorptive theraphy has shown an intermediate level of mechanical properties when compared against the control and immobilized bone samples. These results were found in close agreement with previous results from Dr. Schaffler's research project, which have shown that anti-resorptive treatment with bisphosphonate reduced bone loss in disuse but did not completely inhibit it. Dr. Schaffler data suggest that disuse osteoporosis differs from other osteoporoses in its sensitivity to antiresorptive treatment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Assessment of bone quality, and in particular of osteoporosis
Task Progress:	The results from the present project support this suggestion and demonstrate the ability of ultrasound procedure to assess differences between normal, disuse-induced osteoporotic and normal bone. Moreover, the analysis of our data for the three anatomical directions have shown that beside changes in bone density, the ratio of anisotropy in healthy bone is highly modified during disuse. However, this ratio of anisotropy can be preserved in disuse-induced osteoporotic bone when using the antiresorptive therapy. Observations of directional (anatomical) changes of bone loss undergoing disuse and treated with anti-resorptive therapy were effectively assessed through changes in mechanical properties from ultrasound measurements. Furthermore, mechanical properties obtained from ultrasound measurements were compared against mechanical properties computed from a poroelastic model using measurements from 3D images of the trabecular bone samples from microCT images. In particular, it was found a high correlation between mechanical properties derived from ultrasound measurements and obtained from direct measurement of the microstructure in samples. Finally, using the poroelastic model, the directional information of the microstructure was extracted from each ultrasound measurement and compared to direct measurements from microCT images. The correlation between the bone density obtained from ultrasound and microCT images was moderated, but highly improved when the bone density was directionally modulated with this structural information. The structural information derived from ultrasound measurements was then shown relevant to improve the assessment of mechanical properties when analyzing anisotropic cancellous bone. The results obtained during the first year of the present project suggest that the ultrasound measurements and/or the use of the new poroelastic model can effectively distinguish directional and global changes in trabecular bone undergoing disuse and treated with antiresorptive therapy. The effectiveness of the ultrasound measurements to determine anisotropic changes in mechanical properties of bone was demonstrated against direct analysis of 3D images obtained from microCT. The Aim of the proposed research for the coming year consist in determine microstructure and mechanical properties of bone that has been allowed to recover from long-term disuse (restoration of mechanical loading + bone anabolic agent). In these experiments, anisotropic mechanical and structural properties of cancellous bone will be analyzed from a newly initiated series of experiments (NSBRI grant, Bone recovery after bisphosphonate and PTH treatment of disuse osteoporosis BL00406) that examines recovery potential for disuse and bisphosphonate treated bone, once mechanical loading has been restored. Data suggest that once trabeculae are lost, recovery can only occur through thickening of the remaining trabeculae. These studies will determine whether our acoustical anisotropy approach remains a good predictor of anisotropic mechanical properties in "restored" as well as normal and osteoporotic bone. Anisotropic mechanical properties of bone in which recovery was augmented using a bone anabolic agent (PTH) will be also examined.
Bibliography Type:	Description: (Last Updated: 08/21/2020)
Articles in Peer-reviewed Journals	Cardoso L, Schaffler MB. "Changes of elastic constants and anisotropy patterns in trabecular bone during disuse-induced bone loss assessed by poroelastic ultrasound." J Biomech Eng. 2015 Jan;137(1):0110081–9. https://doi.org/10.1115/1.4029179 ; PMID: 25412022; PMCID: PMC4290507 , Jan-2015
Presentation	Cardoso , L; Seo, S; Jepsen K J; Oddou C; Meunier A; Schaffler, M B "Structurally-based ultrasonic approach of cancellous bone characterization" N/A Feb-2005
Presentation	Cardoso , L; Seo, S; Jepsen K J; Oddou C; Meunier A; Schaffler, M B "Ultrasonic characterization of anisotropic trabecular bone undergoing disuse osteoporosis with and without antiresorptive therapy" N/A Feb-2005
Presentation	Cardoso, L; Seo, S; Jepsen K J; Schaffler, M B "Cardoso, L; Seo, S; Jepsen K J; Schaffler, M B" N/A Jun-2005